

Loot Boxes and Microtransactions in Video Games: A Comparative Legal Analysis

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Introduction

The video gaming industry now functions as a global economic giant which brought in more than \$180 billion during 2021.¹ Game developers expanded their industry by adopting new business models which extended beyond the initial single purchase system. Loot boxes alongside microtransactions have developed into especially lucrative yet intensely debated financial channels among current industry models.² Players acquire digital content through these purchases which provides them access to virtual items and abilities and cosmetic enhancements within games that appear randomly.

Unlike traditional gambling, which is heavily regulated worldwide, loot boxes exist in a legal gray area. Loot boxes contain gambling features yet exist within games marketed to children and teens through the sharing of gambling characteristics which include randomness and real currency usage alongside spending compulsions.³ Due to this lack of regulation many officials such as consumer advocates and parents alongside lawmakers have expressed rising concerns regarding loot box risks and suitable governance.

The regulatory response to loot boxes has varied significantly across different countries and legal systems. While some jurisdictions have taken primary stances of treating loot boxes as gambling and regulating it as such, some have taken more minimalist approaches of requiring transparency and consumer protection, and some have left the regulation mostly in industry self-regulation.⁴

This paper examines how different countries are regulating loot boxes and microtransactions in video games, focusing on their legal status as potential gambling, the various regulatory frameworks being applied, consumer protection measures, as well as the balance between government intervention and industry self-regulation. By comparing different approaches across the United States, European Union, and Asian countries, this analysis aims to identify

¹ Wijman, T. (2023). Free Global Games Market Report. Newzoo. <https://clck.ru/3A9d8c>

² Kelly, M. (2019, February 19). How Loot Boxes Hooked Gamers and Left Regulators Spinning. The Verge. <https://www.theverge.com/2019/2/19/18226852/loot-boxes-gaming-regulation-gambling-free-to-play>

³ Drummond, A., & Sauer, J. D. (2018). Video game loot boxes are psychologically akin to gambling. *Nature Human Behaviour*, 2(8), 530-532.

⁴ Xiao, L. Y. (2021). Regulating loot boxes as gambling? Towards a combined legal and self-regulatory consumer protection approach. *Interactive Entertainment Law Review*, 4(1), 27-47.

emerging best practices as well as potential paths forward that balance consumer protection with the continued innovation of the gaming industry.

Defining Loot Boxes and Microtransactions

Definition and Characteristics

Microtransactions are small purchases made within video games using real-world currency. These purchases can include a wide range of virtual items, from cosmetic alterations with no gameplay impact to powerful items that provide significant advantages.⁵ They have become a standard feature in many games, particularly mobile games and those following the "free-to-play" model where the base game is free but additional content requires payment.

Loot boxes are a specific type of microtransaction that involves an element of chance. Players purchase a virtual container without knowing exactly what items they will receive, similar to buying a lottery ticket or playing a slot machine.⁶ The contents typically vary in rarity and desirability, with more valuable items having lower probabilities of appearing. This randomized reward structure is what has drawn comparisons to gambling and attracted regulatory attention.

According to Nielsen and Grabarczyk's categorization framework, loot boxes can be further classified based on whether players must pay real-world money to access them and whether the rewards have real-world value.⁷ The most controversial type involves both real money payment and rewards with potential real-world value, which most closely resembles traditional gambling.

Evolution in the Gaming Industry

While microtransactions in some form date back to the early 1990s, the modern loot box model gained prominence in Asian free-to-play games before spreading globally around 2010.⁸ The "gacha" mechanics in Japanese mobile games, where players spend money for random character draws, were early predecessors to today's loot boxes.

⁵ Duverge, G. (2016). Insert More Coins: The Psychology Behind Microtransactions. Touro University Worldwide. <http://www.tuw.edu/psychology/psychology-behind-microtransactions/>

⁶ Freedman, A. E. (2019). What Are Loot Boxes? Gaming's Big Controversy Explained. Tom's Guide. <https://www.tomsguide.com/us/what-are-loot-boxes-microtransactions,news-26161.html>

⁷ Nielsen, R. K. L., & Grabarczyk, P. (2019). Are loot boxes gambling? Random reward mechanisms in video games. *Transactions of the Digital Games Research Association*, 4(3), 171-207.

⁸ So, S., & Westland, J. C. (2012). *Red Wired: China's Internet Revolution*. Marshall Cavendish.

Loot boxes became more widely known in Western markets through popular games like Overwatch, FIFA, and Counter-Strike: Global Offensive. The controversy around loot boxes reached its peak with the release of Star Wars Battlefront II in 2017, which received widespread criticism for its aggressive monetization strategy.⁹ The backlash led to increased scrutiny from regulators and the gaming community alike.

Business Model and Revenue Impact

The financial impact of loot boxes and microtransactions on the gaming industry has been substantial. Industry research estimated that loot boxes alone generated \$15 billion in revenue in 2020, with projections suggesting growth to \$20 billion by 2025 without regulatory intervention.¹⁰

Major publishers now derive significant portions of their revenue from these in-game purchases. For example, Electronic Arts reported that its "Ultimate Team" modes, which heavily feature loot box mechanics, generated approximately \$1.62 billion in fiscal year 2021, representing 29% of the company's net revenue.¹¹

The prevalence of loot boxes in popular games is notable. A 2021 study found that 77% of the highest-grossing UK iPhone games contained loot boxes, showing an increase from previous years.¹² This widespread implementation, especially in games accessible to children, has fueled calls for greater regulatory oversight.

Loot Boxes as Gambling: Legal Framework Across Jurisdictions

Elements of Gambling in Traditional Legal Frameworks

Traditional gambling laws typically define gambling as requiring three elements: consideration (paying something of value), chance (an outcome determined at least partly by

⁹ Park, G. (2017, November 18). How a Star Wars video game faced charges that it was promoting gambling. Washington Post. <https://www.washingtonpost.com/news/comic-riffs/wp/2017/11/18/how-a-star-wars-video-game-faced-charges-that-it-was-promoting-gambling/>

¹⁰ Moar, J., & Hunt, N. (2021, March 9). Video Game Loot Boxes to Generate Over \$20 Billion in Revenue by 2025. Juniper Research. [https://www.juniperresearch.com/press/press-releases/video-game-loot-boxes-to-generate-over-\\$20-billion](https://www.juniperresearch.com/press/press-releases/video-game-loot-boxes-to-generate-over-$20-billion)

¹¹ Electronic Arts Inc. (2021). Annual Report for the fiscal year ended March 31, 2021. Retrieved from SEC EDGAR database.

¹² Xiao, L. Y., Henderson, L. L., & Newall, P. W. S. (2022). Loot boxes are more prevalent in United Kingdom video games than previously considered: Updating Zendle et al. (2020). *Addiction*, 117(7), 2109-2111.

randomness), and prize (receiving something of value).¹³ These elements have been established in gambling jurisprudence across many jurisdictions but were developed long before digital gaming existed.

When applying these traditional elements to loot boxes, legal systems face several challenges. First, while the payment element is usually clear when real money is used, the "prize" element becomes more debatable when the rewards are virtual items confined within a game environment. Second, the "chance" element is present in most loot box systems but may be combined with skill-based gameplay in the broader game context.¹⁴

Application to Digital Contexts

Different jurisdictions have taken varying approaches to determining whether loot boxes meet their legal definition of gambling. The key point of contention often centers on whether the virtual items obtained from loot boxes constitute "things of value" under gambling laws.¹⁵

The ability to convert virtual items to real-world money has been a decisive factor in some jurisdictions. Countries like the Netherlands initially deemed loot boxes to be gambling only when the virtual items could be transferred to other players and monetized on secondary markets.¹⁶ However, other jurisdictions like Belgium have adopted broader interpretations, considering the subjective value players place on virtual items sufficient to satisfy the prize element of gambling.

Key Differences in How Countries Define Gambling

Three main approaches have emerged in how countries legally classify loot boxes. The "convertibility" approach focuses on whether loot box rewards can be exchanged for real-world money. The UK Gambling Commission has taken this position, stating that loot boxes constitute gambling only when they contain items that can be cashed out.¹⁷ The "subjective value" approach, exemplified by Belgium, considers the perceived value of virtual items to players as sufficient to

¹³ Champion, W., & Rose, I. N. (2018). *Gaming Law in a Nutshell* (2nd ed.).

¹⁴ King, D. L., & Delfabbro, P. H. (2019). Video game monetization (e.g., 'loot boxes'): A blueprint for practical social responsibility measures. *International Journal of Mental Health and Addiction*, 17(1), 166-179.

¹⁵ Xiao, L. Y. (2022). Which implementations of loot boxes constitute gambling? A UK legal perspective on the potential harms of random reward mechanisms. *International Journal of Mental Health and Addiction*, 20(1), 437-454.

¹⁶ Kansspelautoriteit [The Netherlands Gambling Authority]. (2018). *Onderzoek naar loot boxes: Een buit of een last?* [Study into Loot Boxes: A Treasure or a Burden?].

¹⁷ UK Gambling Commission. (2017). *Virtual currencies, eSports and social gaming — position paper*. <https://www.gamblingcommission.gov.uk/PDF/Virtual-currencies-eSports-and-social-casino-gaming.pdf>

constitute a prize, regardless of convertibility to real money.¹⁸ The "integrated gameplay" approach, as seen in recent Dutch court decisions, examines whether loot boxes are an integral part of a broader game of skill or a standalone gambling feature. In 2022, a Dutch court overturned penalties against Electronic Arts, ruling that FIFA's loot boxes were part of a broader game of skill rather than isolated gambling activities.¹⁹ These differing interpretations have led to a fragmented global regulatory landscape, with the same game potentially being treated as containing illegal gambling in one country while being perfectly legal in another.

Regulatory Approaches in the United States

Federal Oversight and FTC Involvement

In the United States, no federal legislation specifically addresses loot boxes, but the Federal Trade Commission (FTC) has taken an active interest in the issue. In 2019, the FTC held a public workshop titled "Inside the Game: Unlocking the Consumer Issues Surrounding Loot Boxes" to gather information on consumer protection concerns related to loot boxes.²⁰ This workshop brought together industry representatives, consumer advocates, researchers, and government officials to discuss the potential harms and benefits of loot box systems. The fact that the FTC, rather than gambling regulators, took the lead on this issue reflects the U.S. approach of viewing loot boxes primarily through a consumer protection lens rather than a gambling regulatory framework.

Following this workshop, the FTC issued a staff report highlighting concerns about potential deception in how loot boxes are marketed and the particular vulnerability of children to these mechanics. However, the FTC stopped short of recommending specific regulations, instead encouraging industry self-regulation and improved transparency.²¹ This cautious approach is consistent with broader U.S. regulatory philosophy, which often favors industry self-regulation over direct government intervention, particularly for emerging technologies and business models.

¹⁸ Belgian Gaming Commission. (2018). Research Report on Loot Boxes. https://www.gamingcommission.be/opencms/export/sites/default/jhksweb_nl/documents/onderzoeksrapport-loot-boxen-final-publicatie.pdf

¹⁹ Raad van State [Council of State] (Netherlands). (2022). Uitspraak [Ruling] 202005769/1/A3, ECLI:NL:RVS:2022:690.

²⁰ Federal Trade Commission. (2019, August 7). Inside the Game: Unlocking the Consumer Issues Surrounding Loot Boxes. <https://www.ftc.gov/news-events/events-calendar/inside-game-unlocking-consumer-issues-surrounding-loot-boxes>

²¹ Federal Trade Commission. (2020). FTC Staff Presents Report on Loot Boxes to Federal Trade Commissioners. <https://www.ftc.gov/news-events/press-releases/2020/08/ftc-staff-presents-report-loot-boxes-federal-trade-commissioners>

It also reflects the political realities of the U.S. system, where powerful industry lobbying can make it difficult to implement new regulations in the absence of overwhelming public demand.

State-Level Initiatives and Legislation

In the absence of federal action, several states have proposed legislation to address loot boxes. Hawaii was among the first to introduce bills in 2018 that would have prohibited the sale of games containing loot boxes to players under 21 and required disclosure of probability rates.²² However, these bills did not pass. The Hawaii legislation was directly inspired by the Star Wars Battlefront II controversy, demonstrating how high-profile incidents can drive regulatory responses even in the absence of comprehensive research or regulatory frameworks.

California proposed similar legislation (Assembly Bill 2194) in 2018 that would have required games with loot boxes to include a warning label and disclose odds. This bill also failed to advance.²³ Other states, including Washington, Minnesota, and Indiana, have either introduced legislation or commissioned studies on loot boxes, all with limited success thus far. These state-level efforts reflect the traditional role of states as "laboratories of democracy" in the U.S. federal system, experimenting with different regulatory approaches that might eventually influence federal policy. They also highlight the challenge of regulating a global digital industry at the state level, where different requirements across states could create significant compliance challenges for game developers.

In May 2019, a more significant federal legislative effort emerged when Senator Josh Hawley introduced the "Protecting Children from Abusive Games Act," which would have prohibited loot boxes and pay-to-win mechanics in games targeted at children.²⁴ The bill received bipartisan support but ultimately did not progress to become law. This proposed legislation represented the most serious federal attempt to regulate loot boxes to date, specifically framing the issue as one of protecting children rather than addressing gambling more broadly. Its failure to advance despite bipartisan support suggests significant obstacles to federal regulation in this area, potentially including strong industry opposition and competing legislative priorities.

²² H.R. 2686, 29th Leg., Reg. Sess. (Haw. 2018); H.R. 2727, 29th Leg., Reg. Sess. (Haw. 2018).

²³ Assemb. B. 2194, 2017-2018 Leg., Reg. Sess. (Cal. 2018).

²⁴ Protecting Children from Abusive Games Act, S. 1629, 116th Cong. (2019).

Court Cases Addressing Loot Boxes

Loot boxes have not been commonly recognized as gambling by American courts. In cases like *Soto v. Still*, courts have usually stated that virtual items from the loot boxes are not “things of value” under gambling legislation because those items cannot be sold or traded for real money on the game.²⁵ These judicial interpretations are more or less in tune with the convertibility approach that has been observed in other jurisdictions; which demands an immediate redemption option on virtual commodities in order for them to be considered as gambling rewards. This is because the American legal culture a strict textualist approach to legal interpretation as opposed for purposively broader analytical methodologies that could include consideration of psychological effects or valuation.

More recent class action lawsuits have attempted different legal approaches. In 2020, a lawsuit was filed against Electronic Arts in the Northern District of California, alleging that the company's Ultimate Team packs violated California's gambling laws.²⁶ The case was ultimately dismissed due to the plaintiff's acceptance of EA's terms of service from which he or she was barred from suing the company since the company's terms of service included arbitration. This perhaps underlines the challenges that plaintiffs go through in such cases since game firms are usually able to commitments such disputes for arbitration as provided by terms of service. It also illustrates the problem with employing lawsuit instead of legislation for new problems in technologies and consumer rights.

In Canada, a similar class action against Electronic Arts did happen wherein the lawsuit progressed to some extent. In *Sutherland v. Electronic Arts Inc.*, a judge of British Columbia stated that while the gambling claim could not go forward, the consumer protection claim regarding the loot boxes did.²⁷ This means that consumer protection laws may be a better avenue for approaching legal challenges of loot boxes than gambling laws in North America. This is seen in the fact that while courts have found the mechanics of loot boxes to be tantamount to gambling, the specific categorisation of these mechanics as a form of gambling is not as clearly defined, a fact that shows that there is recognition of some distinction between the regulatory regime governing gambling and wider consumer protection laws. It may also suggest that there is a change in direction in how

²⁵ *Soto v. Sky Union*, 159 F. Supp. 3d 871 (N.D. Ill. 2016).

²⁶ *Zajonc v. Electronic Arts, Inc.*, Case No. 20-cv-07871-TSH (N.D. Cal. filed Nov. 9, 2020).

²⁷ *Sutherland v. Electronic Arts Inc.*, 2020 BCSC (Notice of Civil Claim filed September 30, 2020).

such cases will be handled in the future primarily with courts acknowledging that while gambling-based theories may not have merit; consumer protection issues may well be real.

European Union Regulatory Framework

United Kingdom's Position

The UK Gambling Commission has maintained that most loot boxes do not constitute gambling under current UK law because the virtual items obtained cannot be cashed out for real money. In a 2017 position paper, the Commission stated that "where in-game items obtained via loot boxes are confined for use within the game and cannot be cashed out it is unlikely to be caught as a licensable gambling activity."²⁸ This position aligns with the convertibility approach seen in some other jurisdictions, creating a relatively permissive environment for loot boxes that don't facilitate direct cash-out. The UK's stance has been influential globally due to the size of its gaming market and the prominence of its gambling regulatory system.

However, both the Digital, Culture, Media and Sport Committee of the House of Commons and the House of Lords Select Committee on the Social and Economic Impacts of the Gambling Industry have called for loot boxes to be regulated as gambling.²⁹ In response to these calls, the UK government conducted a review but ultimately decided against immediate gambling legislation for loot boxes, citing a lack of evidence for a causal link between loot box spending and problem gambling. This tension between regulatory authorities and legislative committees highlights the competing perspectives on loot boxes within the UK government. It also demonstrates how evidence requirements can shape regulatory outcomes, with the government prioritizing established causal links over precautionary approaches that might address potential harms before they are conclusively proven.

Instead, in July 2023, the UK adopted an industry self-regulation approach with a set of "Industry Principles" to improve player protections. These principles include disclosure of drop rates, targeting unauthorized third-party websites facilitating the sale of items, and a commitment

²⁸ UK Gambling Commission. (2017). Virtual currencies, eSports and social gaming — position paper. <https://www.gamblingcommission.gov.uk/PDF/Virtual-currencies-eSports-and-social-casino-gaming.pdf>

²⁹ House of Commons Digital, Culture, Media and Sport Committee. (2019). Immersive and addictive technologies. <https://publications.parliament.uk/pa/cm201719/cmselect/cmcmums/1846/1846.pdf>

to "lenient" refund policies.³⁰ This approach reflects the UK's general preference for co-regulatory models that combine industry self-governance with government oversight, rather than direct statutory regulation. It remains to be seen whether these voluntary measures will adequately address the concerns raised by parliamentary committees and consumer advocates. The effectiveness of this approach will likely be closely monitored by other jurisdictions considering similar co-regulatory models.

Belgium's Outright Ban

Belgium has taken perhaps the strictest stance on loot boxes globally. In 2018, the Belgian Gaming Commission investigated four popular games (Overwatch, FIFA 18, Counter-Strike: Global Offensive, and Star Wars Battlefront II) and determined that their loot box mechanics violated Belgian gambling law.³¹

The Commission's interpretation focused on the three elements of Belgian gambling law: a game element, a bet leading to profit or loss, and chance playing a role. Notably, Belgium did not require that virtual items be convertible to real money to constitute gambling. The perceived subjective value to players was deemed sufficient to satisfy the prize element.³²

Following this ruling, game publishers faced potential criminal penalties of up to €800,000 and five years imprisonment if they continued to offer paid loot boxes in Belgium. Most major publishers, including Blizzard and 2K Games, complied by either removing loot boxes entirely from the Belgian versions of their games or making them unavailable for purchase with real money.³³

However, recent research has questioned the effectiveness of Belgium's ban. A 2023 study found that despite the legal prohibition, 82% of the 100 highest-grossing iPhone games in Belgium

³⁰ UKIE. (2023). New Principles and Guidance on Paid Loot Boxes. <https://ukie.org.uk/news/new-principles-and-guidance-on-paid-loot-boxes>

³¹ Belgian Gaming Commission. (2018). Research Report on Loot Boxes. https://www.gamingcommission.be/opencms/export/sites/default/jhksweb_nl/documents/onderzoeksrapport-loot-boxen-final-publicatie.pdf

³² Geens, K. (2018, April 25). Loot boxes in three video games in violation of gambling legislation. <https://www.koengeens.be/news/2018/04/25/loot-boxen-in-drie-videogames-in-strijd-met-kansspelwetgeving>

³³ Blizzard Entertainment. (2018). Paid loot boxes and loot chests disabled for players in Belgium. Official Overwatch Forums. <https://eu.forums.blizzard.com/en/overwatch/t/paid-loot-boxes-and-loot-chests-disabled-for-players-in-belgium/8139>

still contained some form of randomized monetization method.³⁴ This suggests significant challenges in enforcement, even with clear legal prohibitions.

Netherlands' Approach

The Netherlands initially took a position somewhat similar to Belgium's but with a crucial difference. In 2018, the Dutch Gaming Authority determined that loot boxes violated gambling laws, but only when the rewards could be transferred to other players and had real-world value.³⁵ This meant that games like FIFA, where players could trade cards on secondary markets, were potentially illegal, while games with untradeable loot box rewards remained legal.

The Dutch Gaming Authority fined Electronic Arts €10 million for violations related to FIFA's Ultimate Team packs. However, EA appealed this decision, and in a landmark ruling in March 2022, the Dutch Council of State overturned the fine. The court found that FIFA's loot boxes were not a standalone gambling activity but part of a broader game of skill.³⁶

This ruling established criteria that would exempt loot boxes from Dutch gambling laws: (a) the loot box mechanic must be part of a broader game; (b) the broader game must be a game of skill; (c) loot boxes must be earned and opened within the game; and (d) loot boxes must mostly be obtainable through organic gameplay rather than purchases.³⁷ This represents a significant retreat from the Netherlands' initial stricter stance.

EU-Wide Considerations

While the European Union has not issued specific regulations on loot boxes, the European Parliament has expressed concerns. In 2020, a report commissioned by the Committee on the Internal Market and Consumer Protection of the European Parliament argued that divergent national regulations on loot boxes could lead to increased compliance costs for companies and contradict the principles of the European Single Market.³⁸

³⁴ Xiao, L. Y. (2023). Breaking Ban: Belgium's ineffective gambling law regulation of video game loot boxes. *Collabra: Psychology*, 9(1), 57641.

³⁵ Kansspelautoriteit [The Netherlands Gambling Authority]. (2018). Onderzoek naar loot boxes: Een buit of een last? [Study into Loot Boxes: A Treasure or a Burden?].

³⁶ Raad van State [Council of State] (Netherlands). (2022). Uitspraak [Ruling] 202005769/1/A3, ECLI:NL:RVS:2022:690.

³⁷ Xiao, L. Y., & Declerck, P. (2022). Video game loot boxes are NOT gambling under Dutch gambling regulation? Shifting the goalpost in *Electronic Arts v Kansspelautoriteit*. OSF Preprints. <https://osf.io/pz24d/>

³⁸ Cerulli-Harms, A., Münsch, M., Thorun, C., Michaelsen, F., & Hausemer, P. (2020). Loot boxes in online games and their effect on consumers, in particular young consumers. Policy Department for Economic, Scientific and Quality of

Spain has recently proposed some of the strictest regulations in the EU, with a draft law that would treat games containing loot boxes similar to gambling. The proposed measures include identity verification for age, restrictions on advertising, publication of probability rates, mandatory self-exclusion systems, and predetermined spending limits.³⁹

Finland has also moved toward regulation, with a bill introduced in 2022 that would amend the definition of "lottery" under the Finnish Lotteries Act to include "virtually utilisable profits," which would encompass loot boxes even when the items cannot be exchanged for real money.⁴⁰

Germany has taken a different approach by focusing on youth protection. In 2021, amendments to the Protection of Young Persons Act led to updated video game classification standards that consider "interaction risks," including loot boxes. This allows the German age classification board (USK) to assign higher age ratings to games with loot boxes, potentially restricting their availability to younger players.⁴¹

Asian Regulatory Models

Japan's Gacha Regulations

Japan was the first country to regulate loot box-like mechanics when its Consumer Affairs Agency declared "complete gacha" (kompu gacha) games illegal in 2012.⁴² Complete gacha refers to a system where players must collect a complete set of items through random draws before receiving a special rare item.

The Agency's decision was based on consumer protection concerns rather than gambling laws, citing "extremely high charges imposed on players" and numerous complaints.⁴³ Notably, many major game developers had already stopped using complete gacha mechanics before the official ban, suggesting that industry self-regulation was partially effective in this case.

Life Policies (EU).

³⁹ Ministerio de Consumo. Anteproyecto de Ley por el que se regulan los mecanismos aleatorios de recompensa asociados a productos de software interactivo de ocio.

⁴⁰ LA 42/2022 vp, Bill to amend Section 2 of the Lotteries Act 2001.

⁴¹ Unterhaltungssoftware Selbstkontrolle. (2022, December 14). In-game purchases, chats and loot boxes: USK expands test criteria.

⁴² Gantayat, A. (2012, May 18). Complete Gacha Officially Deemed Illegal. Andriasang. https://web.archive.org/web/20120521101824/http://andriasang.com/con1kz/complete_gacha_illegal/

⁴³ The Yomiuri Shinbun. (2012, May 6). 'Kompu gacha' online games may be illegal.

Standard gacha mechanics, where players make individual random draws without the complete set requirement, remain legal in Japan but subject to industry self-regulation. After the complete gacha ban, Japanese game companies formed a council to develop self-regulatory guidelines, including the disclosure of probability rates and restrictions on certain manipulative practices.⁴⁴

China's Probability Disclosure Requirements

China has adopted one of the most comprehensive regulatory frameworks for loot boxes, focusing on transparency rather than prohibition. In May 2017, China's Ministry of Culture implemented regulations requiring game publishers to publicly disclose the probabilities of obtaining different items from loot boxes.⁴⁵

The Chinese regulations also include several additional restrictions: (1) loot boxes cannot be purchased with real currency (although this has been circumvented by selling in-game currency with "free" loot boxes included); (2) developers must publicly disclose player spending for the previous 90 days; and (3) games cannot contain "compulsion loops" designed to lead players to use loot boxes, or they may be denied approval by China's game content regulatory body.⁴⁶

Research suggests that the Chinese probability disclosure requirement has been more effective than industry self-regulation in other countries. A study comparing compliance rates found that 95.6% of games in China disclosed probabilities in 2020, compared to only 64.0% of games in the UK under self-regulation in 2021.⁴⁷ However, many of these disclosures were not prominently placed, with only 5.5% of games displaying the information on the purchase page.

South Korea's Approach

South Korea has progressively increased its regulation of loot boxes. Following controversy surrounding the MapleStory game in 2021, where the developer admitted it was

⁴⁴ Schwiddessen, S. (2018). Loot boxes in Japan: Legal analysis and kompu gacha explained. Lexology. <https://www.lexology.com/library/detail.aspx?g=9207df10-a8a2-4f67-81c3-e2f2363df9e6>

⁴⁵ Gartenberg, C. (2017, May 2). China's new law forces Dota, League of Legends, and other games to reveal odds of scoring good loot. The Verge. <https://www.theverge.com/2017/5/2/15517962/china-new-law-dota-league-of-legends-odds-loot-box-random>

⁴⁶ Tang, T. (2018, May 16). A Middle-Ground Approach: How China Regulates Loot Boxes And Gambling Features In Online Games. Mondaq.

⁴⁷ Xiao, L. Y., Henderson, L. L., & Newall, P. (2021). What are the odds? Lower compliance with Western loot box probability disclosure industry self-regulation than Chinese legal regulation. OSF Preprints. <https://osf.io/g5wd9/>

impossible to obtain certain combinations of powerful abilities through loot boxes, the country took stronger regulatory action.⁴⁸

In February 2023, the National Assembly of South Korea passed an amendment to the Game Industry Promotion Act requiring the disclosure of loot box probability rates in-game, on the game's official website, and in advertisements.⁴⁹ Failure to disclose rates, or doing so falsely, is punishable by a fine of up to ₩20 million (\$15,000 USD) or imprisonment for up to two years.

South Korea's approach is notable for combining industry self-regulation with stronger legal requirements. The Korea Game Industry Association had announced its own regulations requiring probability disclosure in 2021, but lawmakers determined that self-regulation was insufficient and proceeded with legislative action.⁵⁰

Consumer Protection Laws and Microtransactions

Existing Consumer Protection Frameworks

Beyond gambling regulations, consumer protection laws provide another avenue for addressing concerns about loot boxes. These laws typically focus on issues such as misleading advertising, unfair contract terms, and the protection of vulnerable consumers, particularly children.

In the European Union, the Unfair Commercial Practices Directive prohibits misleading actions and omissions in business-to-consumer transactions. Some legal scholars have argued that this directive could apply to loot boxes if game companies fail to adequately disclose the odds of winning valuable items or misrepresent the value of potential rewards.⁵¹

In the United States, the FTC has authority under Section 5 of the FTC Act to act against "unfair or deceptive acts or practices."⁵² This could potentially apply to loot boxes if companies make false claims about odds or fail to disclose material information that would affect consumer purchasing decisions.

⁴⁸ Byung-wook, K. (2021, March 9). Game firms under increasing scrutiny over loot box odds. Korea Herald.

⁴⁹ Mi-hee, K. (2023, February 27). Stochastic Item Information Disclosure Act, passed the plenary session of the National Assembly. GameMeca.

⁵⁰ Min-Je, P. (2021, May 29). Game association introduces own loot box disclosure rules. Korea JoongAng Daily.

⁵¹ Xiao, L. Y. (2022). Drafting video game loot box regulation for dummies: a Chinese lesson. *Information & Communications Technology Law*, 31(3).

⁵² 15 U.S.C. § 45(a)(1).

Disclosure Requirements

Disclosure of probability rates for loot box rewards has emerged as a common regulatory approach. As previously discussed, China legally mandates such disclosures, while South Korea has recently enacted similar requirements. In the United States and much of Europe, disclosure is primarily handled through industry self-regulation.

The major mobile app stores now require games with loot boxes to disclose odds. Apple updated its App Store Review Guidelines in 2017 to require that "apps offering loot boxes or other mechanisms that provide randomized virtual items for purchase must disclose the odds of receiving each type of item to customers prior to purchase."⁵³ Google implemented a similar policy for its Play Store in 2019.

Despite these requirements, research suggests that compliance is inconsistent and that many disclosures are difficult for players to access. A study of UK iPhone games found that only 64% of games with loot boxes disclosed probabilities, and many of these disclosures required multiple steps to access.⁵⁴

Conclusion

The regulation of loot boxes in video games represents a complex and evolving challenge at the intersection of gaming, gambling, and consumer protection. Different countries have adopted widely varying approaches, from Belgium's effective ban to China's transparency requirements to the United Kingdom's industry self-regulation model. These divergent responses reflect not only different legal frameworks and regulatory philosophies but also the genuine complexity of applying traditional concepts of gambling and consumer protection to novel digital products.

The evidence suggests that no single regulatory approach has emerged as clearly superior. Belgium's strict prohibition has faced significant enforcement challenges, with many games continuing to offer loot boxes despite the formal ban. China's transparency requirements have achieved higher compliance rates than industry self-regulation in other countries, but questions remain about their effectiveness in changing consumer behavior. The UK's recent shift toward

⁵³ Apple. (2023). App Store Review Guidelines. <https://developer.apple.com/app-store/review/guidelines/>

⁵⁴ Xiao, L. Y., Henderson, L. L., & Newall, P. (2021). What are the odds? Lower compliance with Western loot box probability disclosure industry self-regulation than Chinese legal regulation. OSF Preprints. <https://osf.io/g5wd9/>

industry self-regulation with government oversight represents a promising hybrid approach, though its effectiveness remains to be seen.

What does seem clear is that some form of regulation is necessary to address the legitimate concerns surrounding loot boxes, particularly their impact on vulnerable populations such as children and individuals with gambling problems. Multiple studies have found correlations between loot box spending and problem gambling symptoms, suggesting that these mechanics may pose particular risks to certain individuals. The widespread marketing of games containing loot boxes to children raises additional concerns about normalizing gambling-like behaviors from an early age.

At the same time, regulation must balance these legitimate concerns with the realities of a global digital industry and the benefits that current business models provide to many players and developers. Heavy-handed approaches that fail to consider the complexities of game design or the global nature of the industry may simply push problematic practices into less regulated spaces rather than eliminating them entirely. They may also have unintended consequences for game development and availability, potentially reducing innovation or increasing upfront costs for consumers.

Looking forward, the most promising regulatory approaches are likely to be those that are proportionate, evidence-based, and adaptable to evolving game mechanics. These might include targeted measures addressing specific harms rather than broad prohibitions, enhanced transparency requirements that ensure meaningful information reaches consumers at relevant decision points, stronger protections for children and other vulnerable users, and international coordination to ensure consistent standards across jurisdictions.

The diversity of current regulatory approaches provides an opportunity for comparative evaluation over time, as researchers and policymakers gather data on the effectiveness of different models in reducing harms while preserving beneficial aspects of current business models. This ongoing process of evaluation and adaptation will be essential as the video game industry continues to evolve and new monetization mechanics emerge. By learning from these varied experiences, stakeholders can work toward regulatory frameworks that effectively protect consumers while supporting continued innovation and growth in this dynamic industry.

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